



# Bhabha University, Bhopal (M.P.)

## Scheme of Examination

**Program: Bachelor of Technology Branch: Computer Science and Engineering Session: 2020-21**

**Semester : V**

**Specialization: Computer Science and Engineering**

| S. No . | Subject Code | Subject Name                   | Maximum Marks Allotted |                |                  |            |                      | Total Marks | Contact Hours per week |          |           | Total Credits |
|---------|--------------|--------------------------------|------------------------|----------------|------------------|------------|----------------------|-------------|------------------------|----------|-----------|---------------|
|         |              |                                | Theory                 |                |                  | Practical  |                      |             | L                      | T        | P         |               |
|         |              |                                | End Sem.               | Mid Sem. Exam. | Quiz/ Assignment | End Sem.   | Term work            |             |                        |          |           |               |
|         |              |                                |                        |                |                  |            | Lab Work & Sessional |             |                        |          |           |               |
| 1.      | CS-51        | Theory of Computation          | 70                     | 20             | 10               | 30         | 20                   | 150         | 3                      |          | 2         | 4             |
| 2.      | CS-52        | Database Management Systems    | 70                     | 20             | 10               | 30         | 20                   | 150         | 3                      |          | 2         | 4             |
| 3.      | CS-53        | Cyber Security                 | 70                     | 20             | 10               | -          | -                    | 100         | 3                      | 1        | -         | 4             |
| 4.      | CS-54        | Internet and Web Technology    | 70                     | 20             | 10               | -          | -                    | 100         | 3                      | 1        | -         | 4             |
| 5.      | CS-55        | Computer Graphics & Multimedia | 70                     | 20             | 10               | 30         | 20                   | 150         | 3                      | -        | 2         | 4             |
| 6.      | CS-56        | Lab (Linux)                    | -                      | -              | -                | 30         | 20                   | 50          | -                      | -        | 4         | 2             |
| 7       | CS-57        | Minor Project -1               |                        |                |                  |            | 50                   | 50          |                        |          | 4         | 2             |
|         |              | <b>Total</b>                   | <b>350</b>             | <b>100</b>     | <b>50</b>        | <b>120</b> | <b>130</b>           | <b>750</b>  | <b>15</b>              | <b>2</b> | <b>14</b> | <b>24</b>     |

\* End Semester Examination will be of three hours.

### PRACTICALS SUBJECT :

**CS-51 : Theory of Computation**

**CS-55: Computer Graphics & Multimedia**

**CS-57 : Minor Project-I**

**CS-52: Database Management Systems**

**CS-56 : Lab (Linux)**

|              |               |                |
|--------------|---------------|----------------|
| 1 Hr Lecture | 1 Hr Tutorial | 2 Hr Practical |
| 1 Credit     | 1 Credit      | 1 Credit       |



**BHABHA UNIVERSITY, BHOPAL**

**COURSE CURRICULUM  
FACULTY OF ENGINEERING**

**Undergraduate Programme**

(Courses effective from Academic Year 2020-21)

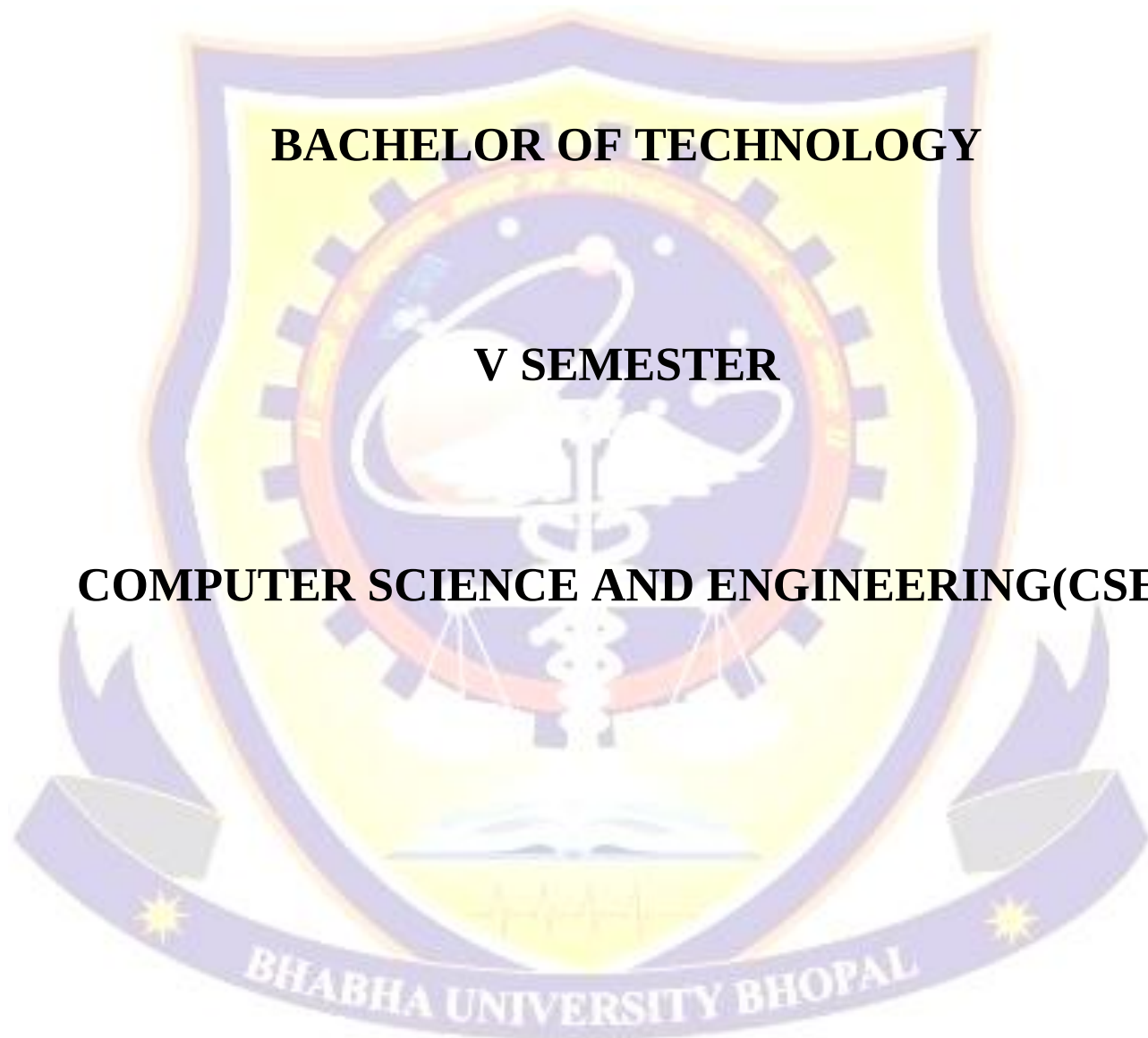
**SYLLABUS OF COURSES TO BE OFFERED  
“Bachelor of Technology”**

**Disclaimer:** The undergraduate programme in Bachelor of Technology syllabus & scheme is uploaded as given by the Faculty concerned to the Academic Council. The same has been approved as it is by the Academic Council on 02.07.2018.

**Dean faculty of Engineering, Bhabha University**

**NH-12, Jatkhedi, Hoshangabad Road, Bhopal (M.P.) – 462026**

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# BHABHA UNIVERSITY, BHOPAL

## CS-51 Theory of Computation

### Unit-I

**Introduction of Automata Theory:** Examples of automata machines, Finite Automata as a language acceptor and translator, Moore machines and mealy machines, composite machine, Conversion from Mealy to Moore and vice versa.

### Unit-II

**Types of Finite Automata:** Non Deterministic Finite Automata (NFA), Deterministic finite automata machines, conversion of NFA to DFA, minimization of automata machines, regular expression, Arden's theorem. Meaning of union, intersection, concatenation and closure, 2 way DFA.

### Unit-III

**Grammars:** Types of grammar, context sensitive grammar, and context free grammar, regular grammar. Derivation trees, ambiguity in grammar, simplification of context free grammar, conversion of grammar to automata machine and vice versa, Chomsky hierarchy of grammar, killing null and unit productions. Chomsky normal form and Greibach normal form.

### Unit-IV

**Push down Automata:** example of PDA, deterministic and non-deterministic PDA, conversion of PDA into context free grammar and vice versa, CFG equivalent to PDA,

### Unit-V

**Turing Machine:** Techniques for construction. Universal Turing machine Multitape, multihead and multidimensional Turing machine, N-P complete problems. Decidability and Recursively Enumerable Languages, decidability, decidable languages, undecidable languages, Halting problem of Turing machine & the post correspondence problem.

### LIST OF EXPERIMENTS

1. Design a Program for creating machine that accepts three consecutive ones.
2. Design a Program for creating machine that accepts the string always ending with 101.
3. Design a program for accepting decimal number divisible by 2.
4. Design a program for creating a machine which accepts string having equal no. of 1's and 0's.
5. Design a program for creating a machine which count number of 1's and 0's in a given string.
6. Design a Program to find 2's complement of a given binary number.
7. Design a Program which will increment the given binary number by 1.
8. Design a Program to convert NFA to DFA..
9. Design a PDA to accept  $WCW^R$  where  $w$  is any string and  $W^R$  is reverse of that string and  $C$  is a Special symbol.
10. Design a Turing machine that accepts the following language  $a^n b^n c^n$  where  $n > 0$ .



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## REFERENCES

- Introduction to Automata Theory Language & Computation, Hopcroft& Ullman, Narosa Publication.
- Element of the Theory Computation, Lewis &Christors,Pearson.
- Theory of Computation, Chandrasekhar & Mishra,PHI.
- Theory of Computation, Wood, Harper &Row.
- Introduction to Computing Theory, Daniel I-A Cohen,Wiley.





# **BHABHA UNIVERSITY, BHOPAL**

## **CS-52 Database Management Systems**

### **Unit I**

DBMS Concepts and architecture Introduction, Database approach v/s Traditional file accessing approach, Advantages, of database systems, Data models, Schemas and instances, Data independence, Data Base Language and interfaces, Overall Database Structure, Functions of DBA and designer, ER data model: Entities and attributes, Entity types, Defining the E-R diagram, Concept of Generalization, Aggregation and Specialization. transforming ER diagram into the tables. Various other data models object oriented data Model, Network data model, and Relational data model, Comparison between the three types of models.

### **Unit II**

Relational Data models: Domains, Tuples, Attributes, Relations, Characteristics of relations, Keys, Key attributes of relation, Relational database, Schemas, Integrity constraints. Referential integrity, Intension and Extension, Relational Query languages: SQL-DDL, DML, integrity constraints, Complex queries, various joins, indexing, triggers, assertions, Relational algebra and relational calculus, Relational algebra operations like select, Project, Join, Division, outer union. Types of relational calculus i.e. Tuple oriented and domain oriented relational calculus and its operations.

### **Unit III**

Data Base Design: Introduction to normalization, Normal forms, Functional dependency, Decomposition, Dependency preservation and lossless join, problems with null valued and dangling tuples, multivalued dependencies. Query Optimization: Introduction, steps of optimization, various algorithms to implement select, project and join operations of relational algebra, optimization methods: heuristic based, cost estimation based.

### **Unit IV**

Transaction Processing Concepts: -Transaction System, Testing of Serializability, Serializability of schedules, conflict & view serializable schedule, recoverability, Recovery from transaction failures. Log based recovery. Checkpoints deadlock handling. Concurrency

Control Techniques: Concurrency Control, locking Techniques for concurrency control, time stamping protocols for concurrency control, validation based protocol, multiple granularity. Multi version schemes, Recovery with concurrent transaction. Introduction to Distributed databases, data mining, data warehousing, Object Technology and DBMS, Comparative study of OODBMS Vs DBMS. Temporal, Deductive, Multimedia, Web & Mobile database.

### **Unit V**

Study of Relational Database Management Systems through Oracle/PL SQL

SQL/MySQL: Architecture, physical files, memory structures, background process. Concept of table spaces, segments, extents and block. Dedicated server, multi threaded server. Distributed database, database links, and snapshot. Data dictionary, dynamic performance view. Security, role management, privilege management, profiles, invoker defined security model. SQL queries, Data extraction from single, multiple tables equi-join, non equi-join, self -join, outer join. Usage of like, any, all, exists, in Special operators. Hierarchical queries, inline queries, flashback queries.



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## LIST OF EXPERIMENTS:

1. Delete duplicate row from the table.
2. Display the alternate row from the table.
3. Delete alternate row from the table.
4. Update multiple rows in using single update statement.
5. Find the third highest paid and third lowest paid salary.
6. Display the 3rd, 4th, 9th rows from the table.
7. Display the ename, which is start with j, k, l or m.
8. Show all employees who were hired the first half of the month.
9. Display the three record in the first row and two records in the second row and one record in the third row in a single sql statements.
10. Write a sql statements for rollback commit and savepoints.
11. Write a pl/sql for select, insert, update and delete statements.
12. Write a pl/sql block to delete a record. If delete operation is successful return 1 else return 0.
13. Display name, hire date of all employees using cursors.
14. Display details of first 5 highly paid employees using cursors.
15. Write a database trigger which fires if you try to insert, update, or delete after 7'o'clock

## REFERENCES

1. Date C J, "An Introduction To Database System", Pearson Educations
2. Korth, Silbertz, Sudarshan, "Fundamental of Database System", McGraw Hill
3. Rob, "Data Base System: Design Implementation & Management", Cengage Learning
4. Elmasri, Navathe, "Fundamentals Of Database Systems", Pearson Educations
5. Atul Kahate, "Introduction to Database Management System", Pearson Educations
6. Oracle 9i Database Administration Fundamental-I, Volume I, Oracle Press, TMH.
7. Paneerselvam, "Data Base Management System", PHI Learning
8. dev.mysql.com
9. www.postgresql.org





# **BHABHA UNIVERSITY, BHOPAL**

## **CS- 53 Cyber Security**

### **UNIT 1**

Introduction of Cyber Crime, Challenges of cyber crime, Classifications of Cybercrimes: E- Mail Spoofing, Spamming, Internet Time Theft, Salami attack/SalamiTechnique,

### **UNIT 2**

Web jacking, Online Frauds, Software Piracy, Computer Network Intrusions, Password Sniffing, Identity Theft, cyber terrorism, Virtual Crime, Perception of cyber criminals: hackers, insurgentsandextremistgroupetc.Webserverwerehacking,sessionhijacking.

### **UNIT 3**

Cyber Crime and Criminal justice: Concept of Cyber Crime and the IT Act, 2000, Hacking, Teenage Web Vandals, Cyber Fraud and Cheating, Defamation, Harassment and E-mail Abuse, Other IT Act Offences, Monetary Penalties, jurisdiction and Cyber Crimes, Nature of Criminality, Strategies to tackle Cyber Crime andTrends.

### **UNIT 4**

The Indian Evidence Act of 1872 v. Information Technology Act, 2000: Status of Electronic Records as Evidence, Proof and Management of Electronic Records; Relevancy, Admissibility and Probative Value of E-Evidence, Proving Digital Signatures, Proof of Electronic Agreements, Proving ElectronicMessages.

### **UNIT 5**

Tools and Methods in Cybercrime: Proxy Servers and Anonymizers, Password Cracking, Key loggers and Spyware, virus and worms, Trojan Horses, Backdoors, DoS and DDoS Attacks, Buffer and Overflow, Attack on Wireless Networks, Phishing : Method of Phishing, PhishingTechniques.

## **REFERENCES**

1. Principles of Cyber crime, Jonathan Clough Cambridge University Press
2. John R. Vacca, Computer Forensics:Computer Crime Scene Investigation, 2nd Edition, Charles River Media,2005
3. Cyber Law Simplified, VivekSood, Pub:TMH.
4. Cyber Security by Nina Godbole, SunitBelapure Pub:Wiley-India
5. Information Warfare: Corporate attack and defense in digital world, William Hutchinson, Mathew Warren,Elsevier.
6. Cyber Laws and IT Protection, Harish Chander,Pub:PHI.



# BHABHA UNIVERSITY, BHOPAL

## CS- 54 Internet and Web Technology

### UNIT 01

Introduction: Concept of WWW, Internet and WWW, HTTP Protocol : Request and Response, Web browser and Web servers, Features of Web 2.0 Web Design: Concepts of effective web design, Web design issues including Browser, Bandwidth and Cache, Display resolution, Look and Feel of the Web site, Page Layout and linking, User centric design, Sitemap, Planning and publishing website, Designing effective navigation.

### UNIT 02

HTML : Basics of HTML, formatting and fonts, commenting code, color, hyperlink, lists, tables, images, forms, XHTML, Meta tags, Character entities, frames and frame sets, Browser architecture and Web site structure. Overview and features of HTML5

### UNIT 03

Style sheets : Need for CSS, introduction to CSS, basic syntax and structure, using CSS, background images, colors and properties, manipulating texts, using fonts, borders and boxes, margins, padding lists, positioning using CSS, CSS2, Overview and features of CSS3 JavaScript : Client side scripting with JavaScript, variables, functions, conditions, loops and repetition, Pop up boxes, Advance JavaScript: Javascript and objects, JavaScript own objects, the DOM and web browser environments, Manipulation using DOM, forms and validations, DHTML : Combining HTML, CSS and Javascript, Events and buttons

### UNIT 04

XML : Introduction to XML, uses of XML, simple XML, XML key components, DTD and Schemas, Using XML with application. Transforming XML using XSL and XSLT PHP: Introduction and basic syntax of PHP, decision and looping with examples, PHP and HTML, Arrays, Functions, Browser control and detection, string, Form processing, Files, Advance Features: Cookies and Sessions, Object Oriented Programming with PHP

### UNIT 05

PHP and MySQL: Basic commands with PHP examples, Connection to server, creating database, selecting a database, listing database, listing table names, creating a table, inserting data, altering tables, queries, deleting database, deleting data and tables, PHP myadmin and database bugs

## REFERENCES

1. Developing Web Applications, Ralph Moseley and M. T. Savaliya, Wiley-India
2. Web Technologies, Black Book, dreamtech Press
3. HTML 5, Black Book, dreamtech Press
4. Web Design, Joel Sklar, Cengage Learning
5. Developing Web Applications in PHP and AJAX, Harwani, McGrawHill
6. Internet and World Wide Web How to program, P.J. Deitel & H.M. Deitel, Pearson



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### CS- 55 Computer Graphics & Multimedia

**Unit-I** Introduction to Raster Scan displays, Pixels, Frame buffer, Vector & Character generation, Random Scan systems, Display devices, Scan Conversion techniques, **Line Drawing:** simple DDA, Bresenham's Algorithm, **Circle Drawing Algorithms:** Midpoint Circle drawing and Bresenham's Algorithm, **Polygon fill algorithm:** Boundary-fill and Flood-fill algorithms

**Unit-II 2-D Transformation:** Translation, Rotation, Scaling, Shearing, Reflection. Inverse Transformation, Homogenous coordinate system, Matrices Transformation, Composite Transformation. **Windowing & Clipping:** World Coordinate System, Screen Coordinate System, Viewing Transformation, Line Clipping & Polygon Clipping Algorithms

**Unit-III 3-D Transformations:** Translation, Rotation and Scaling. **Parallel & Perspective Projection:** Types of Parallel & Perspective Projection, **Hidden Surface elimination:** Depth comparison, Back face detection algorithm, Painter's Algorithm, Z-Buffer Algorithm. Curve generation, Bezier and B-spline methods. **Basic Illumination Model:** Diffuse reflection, Specular reflection, Phong Shading, Gouraud shading, Ray Tracing, Color models like RGB, YIQ, CMY, HSV.

**Unit-IV Multimedia :** Characteristics of a multimedia presentation , Uses of Multimedia, **Text** – Types, Unicode Standard ,text Compression, Text file formats, **Audio-** Components of an audio system,

Digital Audio, Digital Audio processing, Soundcards, Audio file formats, Audio Processing software

, **Video-** Video color spaces, Digital Video, Digital Video processing, Video file formats.

**Unit -V Animation:** Uses of Animation, Principles of Animation, Computer based animation, 3D Animation, Animation file formats, Animation softwares. **Compression:** Lossless/Lossy Compression techniques, Image, Audio & Video Compressions, MPEG Standards ,Multimedia Architecture, Multimedia databases

#### LIST OF EXPERIMENT

- 1 Program to Implement Translation of a Line And Triangle
- 2 Write a program to draw a rectangle using line function
- 3 Write a program to draw a line using DDA's line drawing algorithm
- 4 Write a program to draw a line using Bresenham's line drawing algorithm
- 5 Write a program to draw a circle using equation of circle
- 6 Write a program to fill a circle using Boundary Fill Algorithm
- 7 Write a program to fill a circle using Flood Fill Algorithm
- 8 Program to Draw B-Spline Curve.
- 9 Write a program to translate a triangle about the origin
- 10 Program to Implement Scaling Transformation.



## **BHABHA UNIVERSITY , BHOPAL**

### **COMPUTER SCIENE AND ENGINEERING V SEMESTER**

#### **REFERENCES**

1. Rogers, "Procedural Elements of Computer Graphics", Tata McGrawHill
2. Donald Hearn and M.P. Becker "Computer Graphics" PearsonPub.
3. Parekh "Principles of Multimedia" Tata McGrawHill
4. Maurya, "Computer Graphics with Virtual Reality System " , WileyIndia
5. Pakhira,"Computer Graphics ,Multimedia & Animation",PHIlearning
6. Andleigh, Thakral , "Multimedia System Design " PHILearning





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## CS-56 Linux (Lab)

### Overview of Unix/Linux:-

Concepts, Unix/Linux Installation Process, Hardware Requirements for Unix/Linux, Advantages of Unix/Linux, Reasons for Popularity and Success of Linux/Unix Operating System, Features of Linux/Unix Operating System, Kernel, Kernel Functions,

### The Shell Basic Commands, Shell Programming:-

Shell Variables, Branching Control Structures, Loop-Control Structure, Continue and break Statements, Sleep Command, Debugging Script. Use of Linux as web-server, file server, directory server, application server, DNS server, SMTP server, Firewall, Proxy server.

### File System: -

Definition of File System, Defining Geometry, Disk Controller, Solaris File System, Disk Based File Systems, Network-Based File Systems, Virtual File systems, UFS File System, The Boot Block, The Super Block, The Inode, Tuning File System, Repairing File System.

### Process Control:-

Viewing a Process, Command to display Process, Process Attributes, Process States, Process Fields, PS Commands options, PGREP, PRSTAT, CDE Process Manager, Scheduling Process, Scheduling Priorities, Changing the Priority of a time-sharing process, Killing Process.

### System Security:-

Physical Security, Controlling System Access, Restricted Shells Controlling File Access, File Access Commands, Access Control List (ACLs), Setting ACL Entries, Modifying ACL entries on a file, Deleting ACL entries on a file, Restricting FTP, Securing Super User Access, Restricting Root Access, Monitoring super user Access, TCPWrappers.

### Dynamic Host Configuration Protocol: -

Introduction, DHCP Leased Time, DHCP Scopes, DHCP IP Address, Allocation Types, Planning DHCP Deployment, DHCP Configuration files, Automatic Startup of DHCP Server, Configuration of DHCP Clients, Manually Configuring the DHCP.

### Case Study: -

Installation of Linux, Customization of Linux, Installation of SAMBA, APACHE, TOMCAT, Send MAIL, Postfix, Implementation of DNS, LDAP services, Firewall, Proxy server



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### LIST OF EXPERIMENTS:-

1. To Study basic & User status Unix/Linux Commands.
2. Study & use of commands for performing arithmetic operations with Unix/Linux.
3. Create a file called wlcc.txt with some lines and display how many lines, words and characters are present in that file.
4. Append ten more simple lines to the wlcc.txt file created above and split the appended file into 3 parts. What will be the names of these split files? Display the contents of each of these files. How many lines will be there on the last file?
5. Given two files each of which contains names of students. Create a program to display only those names that are found on both the files.
6. Create a program to find out the i node number of any desired file.
7. Study & use of the Command for changing file permissions.
8. Write a pipeline of commands, which displays on the monitor as well as saves the information about the number of users using the system at present on a file called usere.ux.
9. Execute shell commands through vi editor.
10. Installation, Configuration & Customizations of Unix/Linux.
11. Write a shell script that accepts any number of arguments and prints them in the reverse order.

### REFERENCES

1. Venkatesh Murthy, "Introduction to Unix & Shell", Pearson Edu
2. Forouzan, "Unix & Shell Programming", Cengage Learning
3. Sumitab Das, "Unix Concept & Application", TMH
4. Gopalan, Shivaselvan, "Beginners Guide to Unix", PHI Learning
5. Venkateshwavle, "Linux Programming Tools Unveil'ed", BSPublication.
6. Richard Peterson, "Linux Complete Reference", TMH
7. Richard Peterson, "Unix Complete Reference", TMH



## **CS-57 Minor Project-I**

### **Objectives of the course Minor Project are:**

- ☐ To provide students with a comprehensive experience for applying the knowledge gained so far by studying various courses.
- ☐ To develop an inquiring aptitude and build confidence among students by working on solutions of small industrial problems.
- ☐ To give students an opportunity to do something creative and to assimilate real life work situation in institution.
- ☐ To adapt students for latest development and to handle independently new situations.
- ☐ To develop good expressions power and presentation abilities in students.

The focus of the Minor Project-I is on preparing a working system or some design or understanding of a complex system using system analysis tools and submit it the same in the form of a write up i.e. detail project report. The student should select some real life problems for their project and maintain proper documentation of different stages of project such as need analysis market analysis, concept evaluation, requirement specification, objectives, work plan, analysis, design, implementation and test plan. Each student is required to prepare a project report and present the same at the final examination with a demonstration of the working system.

In Minor Project-I students have to make project in any of the programming language which he/she has learn in previous semesters. Students can make project individually or in a group with two-three students.

Students should make Project in the topics which will be beneficial to them in future also. Students have to Prepare file of the project which he/she will make.